Interventional Mean Embeddings

for Distributional Causal Inference

Houssam Zenati

Gatsby Computational Neuroscience Unit
University College London

London School of Economics
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Means can miss distributional changes

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Different image patterns can have the same mean.

Semi-synthetic OCTMNIST illustration; no rejection in this run.

Zenati and Gretton, NeurIPS 2026.

Interventions change outcome frequencies

The same issue arises for sequences and graphs.

Interventions change outcome frequencies

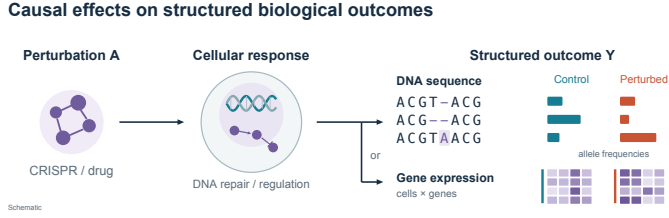
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An average sequence or graph has no canonical meaning without first choosing a representation.

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An average sequence or graph has no canonical meaning without first choosing a representation.



Compare which repaired DNA sequences occur, and how often.

In Repair-seq, perturbing DNA-repair genes changes the frequencies of repaired sequences after a fixed DNA lesion.

Hussmann et al., Cell, 2021.

The distribution is the causal target

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The interventional distribution is a natural causal target on the original outcome space.

Muandet et al., Counterfactual Mean Embeddings, 2021.

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We study population outcome distributions under interventions.

- Testing: detect changes in the interventional distribution.
- Sampling: generate outcomes that represent this distribution.

Scalar distribution functions require an ordering; images and DNA sequences have no natural total order.

Kernel mean embeddings use outcome similarity instead.

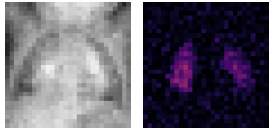
Distributional tests on structured outcomes

Compare interventions q, q' through their outcome laws $P^q, P^{q'}$.

$$H_0 : P^q = P^{q'} \qquad \text{against} \qquad H_1 : P^q \neq P^{q'}.$$

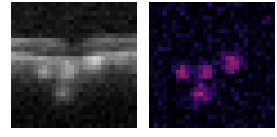
(a) Pneumonia

Treated Y^1 ITE



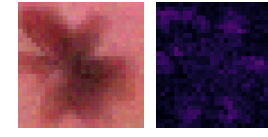
(b) OCT-Drusen

Treated Y^1 ITE



(c) Derma-Melanoma

Treated Y^1 ITE

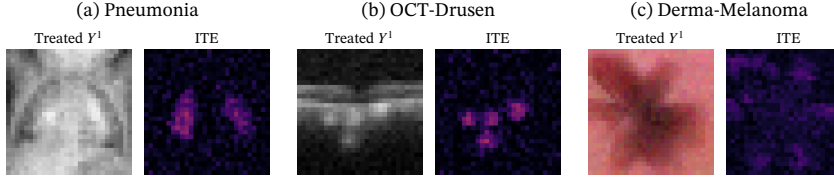


Pattern frequency and severity can change with no mean effect.

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Pattern frequency and severity can change with no mean effect.

The challenge is to detect this change from confounded observations.

Learn deep features that make distributional differences easier to detect.

Semi-synthetic MedMNIST.

Leman et al., submitted to ICLR 2027.

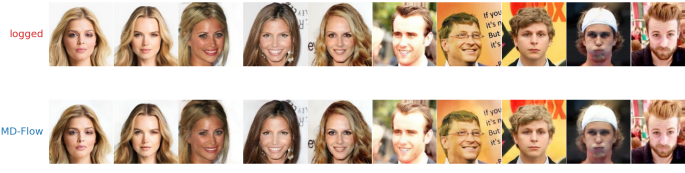
Sampling interventional outcomes

Confounding: treatment assignment and outcomes share causes. The treated group can therefore differ from the target population.

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Generate outcomes as if the intervention were applied to the whole population.

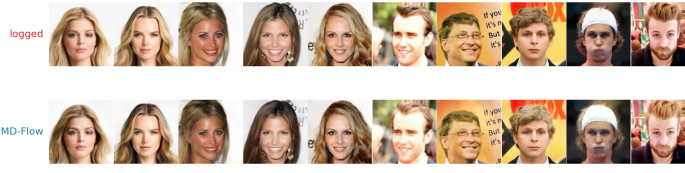


CelebA: match the hair-colour mix across sex groups.

Sampling interventional outcomes

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Generate outcomes as if the intervention were applied to the whole population.



CelebA: match the hair-colour mix across sex groups.

Fairness audit: compare a hair-colour classifier's errors across groups with the same blond/brown proportions.

Population standardisation; classifier audit proposed, not evaluated here.

Bozkurt et al., submitted to AISTATS 2027.

Papers and collaborators

Houssam Zenati, Bariscan Bozkurt and Arthur Gretton. *Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings.* NeurIPS 2025.

Houssam Zenati and Arthur Gretton. *Semiparametric Efficient Test for Interpretable Distributional Treatment Effects.* NeurIPS 2026.

Bariscan Bozkurt, Hugh Dance, Zonghao Chen, Alexandre Galashov, Arthur Gretton and Houssam Zenati. *Sampling Interventional Distributions with Orthogonal MMD Flows.* 2026. Submitted to AISTATS 2027.

Robin Leman, Bariscan Bozkurt, Sofiane Tanji, Arthur Gretton and Houssam Zenati. *Doubly Robust Deep Kernel Tests for Structured Outcomes.* 2026. Submitted to ICLR 2027.

With thanks to my main collaborators



Arthur Gretton



Bariscan Bozkurt



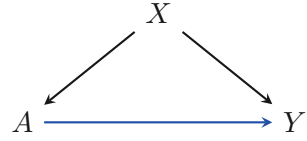
Robin Leman

and also Hugh Dance, Zonghao (Hudson) Chen, Alexandre Galashov and Sofiane Tanji.

The causal graph

At population level, let $O = (X, A, Y) \sim P_0$.

X: pre-treatment confounders; A: treatment; Y: outcome. P_0: observational joint distribution; P_X: its marginal law of X.



The common causes X affect both treatment and outcome.

An intervention changes how treatment is assigned, while retaining the population distribution of X.

Identification requires adjustment for these common causes.

The interventional outcome law

Hard intervention: set $A = a$. Its outcome law is $P^a = \mathcal{L}(Y(a))$.

Soft intervention: replace the treatment mechanism $e_0(da \mid x) = P_0(A \in da \mid X = x)$ by a specified $q(da \mid x)$.

$e_0(a \mid x)$: propensity mass/density; $Y(a)$: outcome under a; $\mathcal{L}$: probability law.

Identification from the observational distribution

$$P^q(dy) = \int\int P_0(dy \mid a, x)\, q(da \mid x)\, P_X(dx).$$

- Consistency: $Y = Y(A)$.
- No unmeasured confounding: $Y(a) \perp A \mid X$.
- Overlap: $q(\cdot \mid x) \ll e_0(\cdot \mid x)$.

Overlap means assigning only supported treatments. For discrete A, setting $q = \delta_a$ recovers P^a; δ_a: point mass at a.

Kernels compare structured outcomes

A positive-definite kernel $k(y, v)$ measures similarity between an outcome y and a reference outcome v.

- Images: compare spatial patterns.
- DNA, protein sequences and text: compare subsequences or alignments.
- Graphs: compare common substructures.

The feature map is a similarity profile in the reproducing kernel Hilbert space (RKHS) $\mathcal{H}$:

$$\phi(y) = k(y, \cdot) \in \mathcal{H}.$$

$\langle f, g \rangle_{\mathcal{H}}$: RKHS inner product; $\|f\|_{\mathcal{H}}^2 = \langle f, f \rangle_{\mathcal{H}}$.

Images: Lazebnik et al., 2006; strings and graphs: Haussler, 1999; biological sequences: Saigo, Vert et al., 2004.

Kernel mean embeddings and the witness

For an outcome law P, such as the observed outcome distribution, define its kernel mean embedding (KME):

$$\mu_P = \mathbb{E}_{Y\sim P}[\phi(Y)], \qquad \mu_P(v) = \mathbb{E}_{Y\sim P}[k(Y, v)].$$

The KME records average similarity to every reference outcome v.

To compare two laws P, P', define their witness function:

$$w(v) = \mu_P(v) - \mu_{P'}(v).$$

A positive value means greater average similarity to v under P.

Bounded measurable kernel; existence conditions in backup.

Smola et al., 2007; Gretton et al., 2012.

Interventional mean embeddings

Define the interventional mean embedding (IME):

$$\mu_q = \mathbb{E}_{Y \sim P^q}[\phi(Y)], \qquad \mu_q(v) = \mathbb{E}_{Y \sim P^q}[k(Y, v)].$$

The averaging law is now interventional, rather than observed.

Its interventional witness compares two interventions:

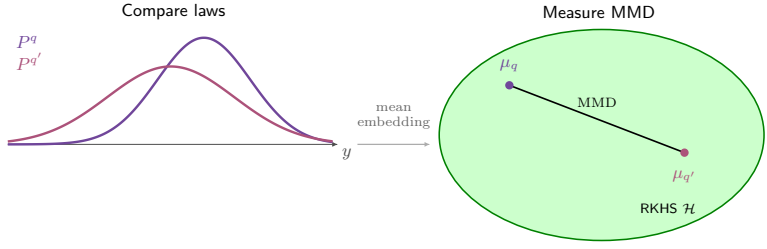
$$w(v) = \mu_q(v) - \mu_{q'}(v).$$

With a characteristic kernel, the embedding identifies the law:

$$\mu_q = \mu_{q'} \quad \Longleftrightarrow \quad P^q = P^{q'}.$$

Muandet et al., Counterfactual Mean Embeddings, 2021.

MMD compares interventional laws



$$\mathrm{MMD}(P^q, P^{q'}) = \|\mu_q - \mu_{q'}\|_{\mathcal{H}}.$$

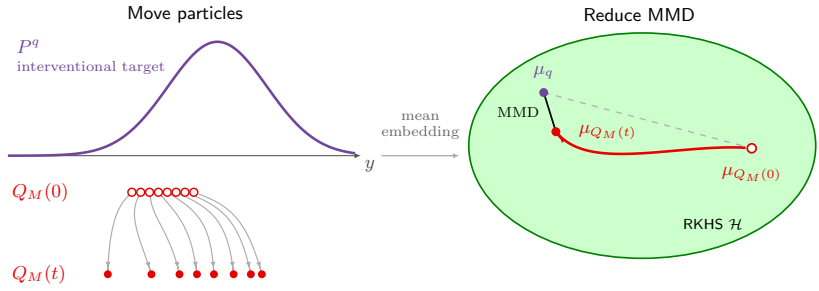
MMD: maximum mean discrepancy. Equal laws give zero distance.

This distance also defines a sampling objective.

Characteristic kernel; schematic outcome densities.

Gretton et al., 2012.

MMD flow transports particles



Move the particles so their embedding approaches the target IME.

Transport the same M particles by updating all their positions. Their number and equal weights stay fixed.

$Q_M(t)$: particle law; $\mu_{Q_M(t)}$: its embedding. Smooth kernel; projected RKHS trajectory.

Arbel et al., NeurIPS 2019.

IME estimation

Causal adjustment uses nuisance functions

From n independent observations $O_i \sim P_0$:

Observed KME: directly average the features.

$$\widehat{\mu}_{P_Y} = \frac{1}{n}\sum_i \phi(Y_i), \qquad \|\widehat{\mu}_{P_Y} - \mu_{P_Y}\|_{\mathcal{H}} = O_p(n^{-1/2}).$$

P_Y: observed outcome law; $\mathbb{E}_0 = \mathbb{E}_{P_0}$; $\mathbb{E}_X = \mathbb{E}_{P_X}$. Bounded kernel; overlap; $O_p(n^{-1/2})$: root-n stochastic error.

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IME: first adjust for the covariates.

$$\mu_q = \mathbb{E}_X\left[\int m_0(a, X) q(da \mid X)\right] = \mathbb{E}_0\left[\frac{q(A \mid X)}{e_0(A \mid X)}\phi(Y)\right].$$

The CME $m_0(a, x) = \mathbb{E}_0[\phi(Y) \mid A = a, X = x]$ and propensity e_0 are nuisances: auxiliary functions needed for our target.

Estimating them can introduce bias and slow the rate.

P_Y: observed outcome law; $\mathbb{E}_0 = \mathbb{E}_{P_0}$; $\mathbb{E}_X = \mathbb{E}_{P_X}$. Bounded kernel; overlap; $O_p(n^{-1/2})$: root-n stochastic error.

Estimating an IME by prediction

The outcome nuisance is the conditional mean embedding:

$$m_0(a, x) = \mathbb{E}_0[\phi(Y) \mid A = a, X = x].$$

Average a candidate model m over the intervention:

$$m_q(x) = \int m(a, x) q(da \mid x), \qquad \mu_q = \mathbb{E}_X[m_{0,q}(X)].$$

A fitted nuisance $\widehat{m}$ gives the plug-in estimator

$$\widehat{\mu}_q^{\mathrm{PI}} = \frac{1}{n} \sum_i \widehat{m}_q(X_i).$$

Nonparametric CME estimation can be slow; the plug-in estimator generally inherits its error.

Subscript 0: truth; hats: fitted quantities.

Zenati et al., NeurIPS 2025.

Estimating an IME by reweighting

The treatment propensity e_0 gives the weight nuisance r_0:

$$r_0(A, X) = \frac{q(A \mid X)}{e_0(A \mid X)}, \qquad \mu_q = \mathbb{E}_0[r_0(A, X)\phi(Y)].$$

For a discrete treatment set to a,

$$r_0(A, X) = \frac{\mathbf{1}\{A = a\}}{e_0(a \mid X)}.$$

A fitted propensity gives the nuisance $\widehat{r} = q/\widehat{e}$

$$\widehat{\mu}_q^{\mathrm{IPW}} = \frac{1}{n} \sum_i \widehat{r}(A_i, X_i)\phi(Y_i).$$

These weights also let us correct errors in the outcome prediction.

Use conditional treatment masses or densities; overlap is required.

Zenati et al., NeurIPS 2025.

Prediction plus a residual correction

Combine the two nuisance models: $\eta = (m, r)$.

m: conditional mean embedding model; r: intervention weight model; $\eta_0 = (m_0, r_0)$: truth.

The feature residual is

$$\phi(Y) - m(A, X).$$

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

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The feature residual is

$$\phi(Y) - m(A, X).$$

Each observation contributes a corrected profile $\xi_q(O; \eta) \in \mathcal{H}$:

$$\xi_q(O; \eta) = \underbrace{m_q(X)}_{\text{prediction}} + \underbrace{r(A, X)}_{\text{assignment adjustment}} \underbrace{\{\phi(Y) - m(A, X)\}}_{\text{residual}}.$$

$$\widehat{\mu}_q^{\mathrm{DR}} = \frac{1}{n} \sum_i \widehat{\xi}_q(O_i).$$

$\widehat{\xi}_q(O_i) = \xi_q(O_i; \widehat{\eta}_{-i})$: use models fitted outside observation i's fold.

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

Either correct model removes population bias

$$\xi_q(O;\eta) = m_q(X) + r(A,X)\{\phi(Y) - m(A,X)\}.$$

Correct outcome model: residuals average to zero:

$$\mathbb{E}_0[\phi(Y) - m_0(A,X) \mid A, X] = 0.$$

The prediction already targets μ_q.

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Correct weights: the correction cancels prediction error:

$$\mathbb{E}_0[r_0 m(A,X)] = \mathbb{E}_X[m_q(X)].$$

Hence $\mathbb{E}_0\xi_q(O;(m,r_0)) = \mathbb{E}_0[r_0\phi(Y)] = \mu_q$.

Either correct model removes the population bias.

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

The nuisance errors multiply

Average the corrected profile at the population level:

$$h_\eta = \mathbb{E}_0[\xi_q(O;\eta)], \qquad h_{\eta_0} = \mu_q, \quad \eta_0 = (m_0, r_0).$$

The two nuisance errors multiply:

$$h_\eta - \mu_q = -\mathbb{E}_0[(r - r_0)(m - m_0)], \qquad \|h_\eta - \mu_q\|_{\mathcal{H}} \le \rho_r \rho_m.$$

Weight error: $\rho_r = \|r - r_0\|_{L^2}$.

CME error: $\rho_m = \|m - m_0\|_{L^2(\mathcal{H})}$.

The correction removes each nuisance's first-order bias.

$\|f\|_{L^2}^2 = \mathbb{E}_0|f(A,X)|^2; \|g\|_{L^2(\mathcal{H})}^2 = \mathbb{E}_0\|g(A,X)\|_{\mathcal{H}}^2.$

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

DR recovers the root-n rate

For the fitted nuisance errors ρ_r, ρ_m, under regularity:

$$\|\widehat{\mu}_q^{\mathrm{DR}} - \mu_q\|_{\mathcal{H}} = O_p\left(\underbrace{n^{-1/2}}_{\text{sampling noise}} + \underbrace{\rho_r \rho_m}_{\text{nuisance bias}}\right).$$

The nuisances can converge slowly, provided their product is small.

If $\rho_r \rho_m = o_p(n^{-1/2})$, nuisance bias is negligible relative to sampling noise.

We recover the root-n scale of the ordinary empirical KME.

Cross-fitting, controlled score second moments and consistency. $O_p(a_n)$: order a_n in probability; $o_p(a_n)$: negligible relative to a_n.

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

IME error controls comparison error

Compare the two interventional embeddings directly. Here $\widehat{\mu}_q$ and $\widehat{\mu}_{q'}$ are their DR estimates.

Comparisons inherit estimation accuracy

$$\begin{aligned}\left|\|\widehat{\mu}_q - \widehat{\mu}_{q'}\|_{\mathcal{H}} - \|\mu_q - \mu_{q'}\|_{\mathcal{H}}\right| \\ \leq \|\widehat{\mu}_q - \mu_q\|_{\mathcal{H}} + \|\widehat{\mu}_{q'} - \mu_{q'}\|_{\mathcal{H}}.\end{aligned}$$

- Interpretable testing: IME evaluations give causal contrasts at selected reference outcomes.
- Sampling: the estimated IME specifies the distribution-matching objective.

We first use these properties to compare two interventional laws.

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

Global testing

Global distributional testing

$$H_0 : P^q = P^{q'}, \qquad H_1 : P^q \neq P^{q'}.$$

Global: compare expectations in all RKHS directions. With a characteristic kernel,

$$H_0 \iff \mu_q = \mu_{q'} \iff \mathrm{MMD}^2 = 0.$$

Subtract the two fitted corrected profiles for each observation:

$$\widehat{d}_i = \widehat{\xi}_q(O_i) - \widehat{\xi}_{q'}(O_i).$$

$\widehat{d}_i$: corrected profile difference; its average estimates $\mu_q - \mu_{q'}$.

Zenati et al., NeurIPS 2025.

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At the null, squared MMD fluctuates quadratically.

The usual first-order Gaussian approximation does not apply. Degeneracy explained in backup.

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Global testing can also use deep features learned for test power.

Zenati et al., NeurIPS 2025.
Leman et al., submitted to ICLR 2027.

Cross-MMD gives Gaussian calibration

1. Fit. Fit nuisances separately. On evaluation splits I_1, I_2, compute $\widehat{d}_i = \widehat{\xi}_q(O_i) - \widehat{\xi}_{q'}(O_i)$.

2. Project. Use I_2 to choose a direction; project I_1 onto it:

$$\bar{d}_2 = \frac{1}{|I_2|} \sum_{j \in I_2} \widehat{d}_j, \qquad f_i = \langle \widehat{d}_i, \bar{d}_2 \rangle_{\mathcal{H}} \quad (i \in I_1).$$

3. Test. With $\bar{f}, s_f$ the mean and standard deviation on I_1,

$$T = \frac{\sqrt{|I_1|}\,\bar{f}}{s_f} \xrightarrow[H_0]{d} \mathcal{N}(0,1), \qquad \text{reject if } T > z_{1-\alpha}.$$

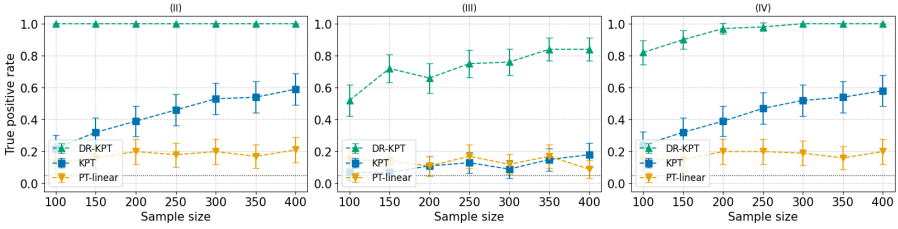
DR makes nuisance bias negligible at the testing scale.

Splitting and studentisation give the Gaussian calibration.

Mean-square consistent scores; product bias $o_p(n^{-1/2})$ for both targets; finite fourth moments and nondegenerate null covariance. $z_{1-\alpha}$: normal quantile; $\xrightarrow{d}$: distributional limit.

Zenati et al., NeurIPS 2025; Kim and Ramdas, Bernoulli 2024.

Effects beyond a mean shift



A mean-preserving mixture still changes the outcome law.

A semi-synthetic image-distribution comparison (dSprites):

RBF method	Null rejection	Power
Reweighted baseline	0.401	0.054
Doubly robust test	0.024	0.656

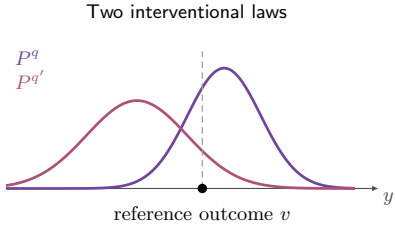
Zenati et al., NeurIPS 2025.

Interpretable testing

Interpretable comparisons

A global test detects a difference. We also want to identify which outcome patterns reveal it.

Each v defines a comparison $k(Y, v)$.

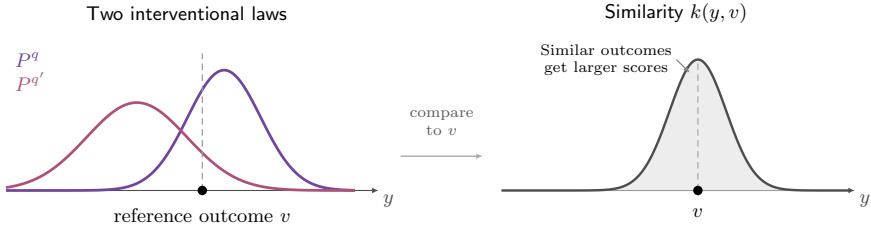


Zenati and Gretton, NeurIPS 2026.

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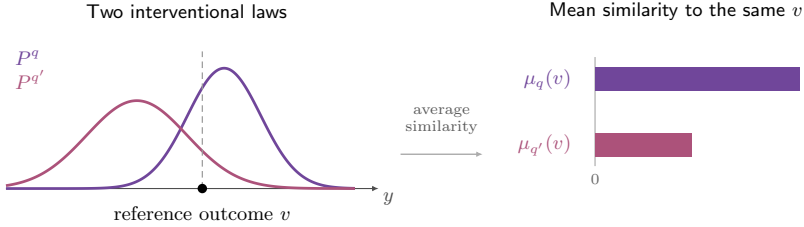
Kernel similarity $k(y, v)$ is large when y resembles v.

Zenati and Gretton, NeurIPS 2026.

The interventional witness

Average $k(Y, v)$ under each intervention:

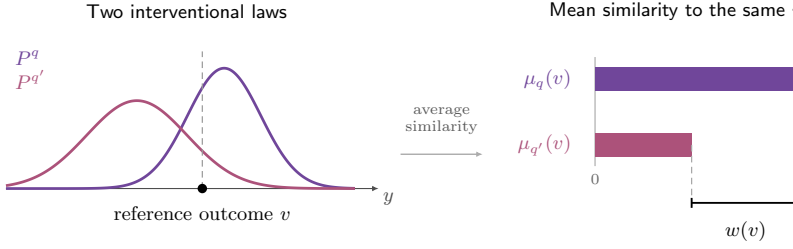
$$\mu_q(v) = \mathbb{E}_{Y \sim P^q}[k(Y, v)], \qquad \mu_{q'}(v) = \mathbb{E}_{Y \sim P^{q'}}[k(Y, v)].$$



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$$w(v) = \mu_q(v) - \mu_{q'}(v).$$

$w(v) > 0$: greater average similarity to v under q.

Finite-location causal contrasts

Collect the witness values at J reference outcomes $V = (v_1, \ldots, v_J)$:

$$\mathbf{w}(V) = \left(\mu_q(v_j) - \mu_{q'}(v_j)\right)_{j=1}^{J}.$$

Each coordinate compares the same outcome pattern under the two interventions.

Equal interventional laws imply $\mathbf{w}(V) = 0$, so we test these contrasts jointly.

Observed treatment groups can differ through confounding. The DR IMEs provide the adjusted comparisons.

A finite set of locations can miss some distributional differences.

Zenati and Gretton, NeurIPS 2026.

DR controls bias in the test

Apply the DR IME correction to the scalar responses $k(Y, v_j)$:

$$\widehat{d}_i(v_j) = \widehat{\xi}_q(O_i)(v_j) - \widehat{\xi}_{q'}(O_i)(v_j), \qquad \widehat{w}_j = \frac{1}{n}\sum_i \widehat{d}_i(v_j).$$

Collect $\widehat{\mathbf{w}}(V) = (\widehat{w}_j)_{j=1}^J$. The sample covariance $\widehat{\Sigma}(V)$ records the variability and dependence of the corrected coordinates.

Nuisance bias can look like an intervention effect. DR bounds this bias by a product of regression and weight errors.

When this bias is negligible at the test's scale, false-positive rates stay calibrated.

Product bias $o_p(n^{-1/2})$; independent nuisance fits or cross-fitting and the earlier regularity conditions. The mean's asymptotic covariance is $\Sigma(V)/n$.

Zenati and Gretton, NeurIPS 2026.

Testing the joint contrast

For fixed locations V, measure the estimated contrast relative to its uncertainty:

$$T_n(V) = n\widehat{\mathbf{w}}(V)^\top \big(\widehat{\Sigma}(V) + \gamma_n I_J\big)^{-1} \widehat{\mathbf{w}}(V).$$

Under the null, $T_n(V) \Rightarrow \chi^2_J$. Reject above the threshold $\chi^2_{J,1-\alpha}$.

At population level, the corresponding criterion is

$$D_P(V) = \mathbf{w}_P(V)^\top \Sigma_P(V)^{-1} \mathbf{w}_P(V).$$

Location learning seeks contrasts that the test can detect.

P: observed-data law; $\mathbf{w}_P$ and Σ_P are its true contrast and corrected-coordinate covariance. Assume $\Sigma_P(V) \succ 0$; the earlier inference conditions and $\gamma_n \to 0$ apply.

Zenati and Gretton, NeurIPS 2026.

Power guides location learning

Effects of size $n^{-1/2}$ match the estimation noise. Along local laws P_n approaching the null,

$$nD_{P_n}(V) \longrightarrow \lambda(V).$$

The same limit determines how often the test rejects:

$$T_n(V) \Rightarrow \chi^2_J(\lambda(V)).$$

At fixed J and level α, larger $\lambda(V)$ means greater power.

DR attains the efficient covariance for these contrasts. This can reveal smaller effects with the same data and false-positive rate.

Efficiency is among regular estimators of fixed contrasts. Regular local alternatives, nonsingular covariance and the earlier inference conditions. Precise local-power limit.

Zenati and Gretton, NeurIPS 2026.

Learning the test locations

Fix the kernel and J. Use three disjoint samples.

Fit. Estimate the nuisance functions on the first sample. Hold these fits fixed.

Learn. On the learning sample, score each candidate set V:

$$\widehat{D}_{\tau,\mathrm{tr}}(V) = \widehat{\mathbf{w}}_{\mathrm{tr}}(V)^{\top}\big(\widehat{\Sigma}_{\mathrm{tr}}(V) + \tau I_J\big)^{-1}\widehat{\mathbf{w}}_{\mathrm{tr}}(V).$$

Select $\widehat{V} \approx \arg\max_V \widehat{D}_{\tau,\mathrm{tr}}(V)$.

Test. Freeze $\widehat{V}$ and use the untouched test sample.

Errors in the contrasts or covariance can mis-rank candidate sets.

tr: learning data; $\tau > 0$: training ridge. Search over continuous locations or a dictionary of valid outcomes.

Zenati and Gretton, NeurIPS 2026.

Testing on held-out data

Location learning can select differences due to sampling noise. Independent test data protect against selection bias.

At the frozen $\widehat{V}$, recompute $\widehat{\mathbf{w}}_{\mathrm{te}}$ and $\widehat{\Sigma}_{\mathrm{te}}$ on n_{te} held-out observations. Under the null,

$$T_{\mathrm{te}} = n_{\mathrm{te}} \widehat{\mathbf{w}}_{\mathrm{te}}^{\top} (\widehat{\Sigma}_{\mathrm{te}} + \gamma_{n_{\mathrm{te}}} I_J)^{-1} \widehat{\mathbf{w}}_{\mathrm{te}} \Rightarrow \chi^2_J.$$

Reject when $p = 1 - F_{\chi^2_J}(T_{\mathrm{te}}) \leq \alpha$.

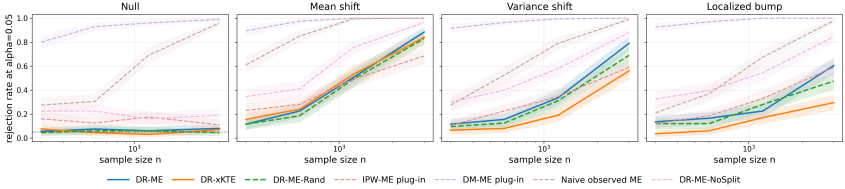
Report the locations, signed contrasts, standard errors and joint p-value:

$$\widehat{\mathrm{se}}(\widehat{w}_{\mathrm{te},j}) = \sqrt{\widehat{\Sigma}_{\mathrm{te},jj}/n_{\mathrm{te}}}.$$

te: test data. Conditional on fitting and selection: fixed J, consistent nondegenerate covariance, score consistency, product bias $o_p(n_{\mathrm{te}}^{-1/2})$ and vanishing test ridge.

Zenati and Gretton, NeurIPS 2026.

Learned locations improve test power



Same DR test; different location rules ($d_Y = 25$):

Location criterion	Power
Full covariance whitening	0.760
Raw witness	0.380
Random locations	0.295

Synthetic benchmarks, level 0.05; plot and table use separate protocols.

Zenati and Gretton, NeurIPS 2026.

Sampling

The estimated IME guides sampling

We can estimate μ_q from observational data, even without samples from the interventional law P^q.

Approximate the target law by transporting M particles:

$$Q_M = \frac{1}{M} \sum_{j=1}^{M} \delta_{z_j}.$$

- Initialise the particles from observed outcomes.
- Move their positions so that Q_M approximates P^q.
- Keep M, the weights $1/M$ and the estimated IME fixed.

The IME specifies the matching objective and the particle velocities.

δ_z: point mass. The output is an empirical distribution; its particles need not be independent. Images use continuous encoder coordinates.

Bozkurt et al., submitted to AISTATS 2027.

Matching the interventional distribution

Compare a candidate distribution Q with the estimated IME:

$$\widehat{F}(Q) = \tfrac{1}{2}\|\mu_Q - \widehat{\mu}_q^{\mathrm{DR}}\|_{\mathcal{H}}^2.$$

With the exact IME, this is $\frac{1}{2}\,\mathrm{MMD}^2(Q, P^q)$.

C is independent of Q_M. A characteristic kernel makes zero exact-target MMD identify P^q; reaching zero is a separate optimisation question.

Bozkurt et al., submitted to AISTATS 2027.

Matching the interventional distribution

Compare a candidate distribution Q with the estimated IME:

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With the exact IME, this is $\frac{1}{2}\,\mathrm{MMD}^2(Q, P^q)$. For $Q_M = M^{-1}\sum_j \delta_{z_j}$, the criterion becomes

$$\widehat{F}(Q_M) = \underbrace{\frac{1}{2M^2}\sum_{j,\ell} k(z_j, z_\ell)}_{\text{particle interactions}} - \underbrace{\frac{1}{M}\sum_j \widehat{\mu}_q^{\mathrm{DR}}(z_j)}_{\text{agreement with the target}} + C.$$

The IME supplies the entire target-dependent part.

The pairwise term compares particles with each other; the target term compares them with the estimated IME.

C is independent of Q_M. A characteristic kernel makes zero exact-target MMD identify P^q; reaching zero is a separate optimisation question.

Bozkurt et al., submitted to AISTATS 2027.

MMD sampling by particle transport

Fixed target: $\widehat{\mu}_q^{\mathrm{DR}}(z) = \sum_i c_i k(z, \bar{z}_i)$.

Differentiable kernel k; coefficients c_i and support points $\bar{z}_i$ stay fixed.

Euler transport

Initialise $z_1^{(0)}, \ldots, z_M^{(0)}$ from observed outcomes.

For $r = 0, \ldots, R-1$, update all particles simultaneously:

$$v_j^{(r)} = \sum_i c_i \nabla_1 k(z_j^{(r)}, \bar{z}_i) - \frac{1}{M} \sum_{\ell=1}^{M} \nabla_1 k(z_j^{(r)}, z_\ell^{(r)}),$$

$$z_j^{(r+1)} = z_j^{(r)} + \varepsilon_r v_j^{(r)}, \qquad j = 1, \ldots, M.$$

Return the empirical distribution $Q_M^{(R)}$.

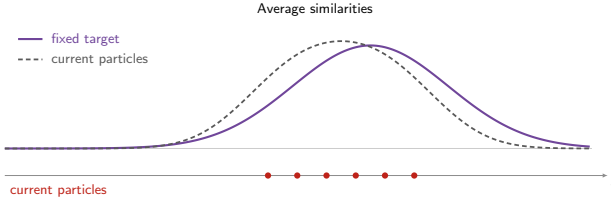
The same M particles move; their weights stay fixed.

∇_1: derivative in the first argument; $\varepsilon_r > 0$: step size. Target coefficients in backup.

Continuous transport decreases MMD

Follow the same particles through one transport step.

Compare the particle and target embeddings.



$$F(Q_M) = \tfrac{1}{2}\|\mu_{Q_M} - \mu_q\|^2_{\mathcal{H}}.$$

Gaussian-kernel illustration. The descent identity holds for smooth kernels in continuous time; discrete updates need a suitable step size.

Arbel et al., NeurIPS 2019. Bozkurt et al., submitted to AISTATS 2027.

Continuous transport decreases MMD

Follow the same particles through one transport step.

The witness gradient determines particle motion.



$$f_r(z) = \mu_{Q_M^{(r)}}(z) - \mu_q(z), \qquad v_j^{(r)} = -f_r'(z_j^{(r)}).$$

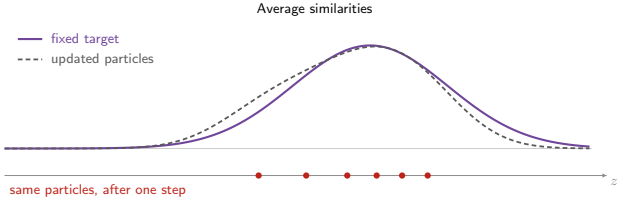
Gaussian-kernel illustration. The descent identity holds for smooth kernels in continuous time; discrete updates need a suitable step size.

Arbel et al., NeurIPS 2019. Bozkurt et al., submitted to AISTATS 2027.

Continuous transport decreases MMD

Follow the same particles through one transport step.

Move the particles and recompute the discrepancy.



$$\frac{d}{dt}F(Q_M(t)) = -\frac{1}{M}\sum_j \|v_j(t)\|^2 \le 0.$$

Gaussian-kernel illustration. The descent identity holds for smooth kernels in continuous time; discrete updates need a suitable step size.

Arbel et al., NeurIPS 2019. Bozkurt et al., submitted to AISTATS 2027.

Sampling error includes IME error

Assess $Q_M^{(R)}$ against the true interventional law P^q:

$$\mathrm{MMD}_k(Q_M^{(R)}, P^q) \leq \underbrace{\sqrt{2\widehat{F}(Q_M^{(R)})}}_{\text{remaining particle discrepancy}} + \underbrace{e_n}_{\text{IME error}} .$$

DR estimation controls the target error:

$$e_n = \|\widehat{\mu}_q^{\mathrm{DR}} - \mu_q\|_{\mathcal{H}} = O_p\Big(n^{-1/2} + \rho_m \rho_r\Big) .$$

DR cancels first-order nuisance bias in the target term.

ρ_m, ρ_r: nuisance errors. The rate uses the earlier estimation assumptions; numerical target errors enter e_n separately.

Iteration and particle guarantee Orthogonality proof.

Bozkurt et al., submitted to AISTATS 2027.

Evaluating sampling on held-out data

Use a fixed kernel and an independently estimated target:

$$\widehat{\mathrm{MMD}}_{\mathrm{eval}} = \|\mu_{Q_M^{(R)}} - \widehat{\mu}_q^{\mathrm{eval}}\|_{\mathcal{H}}.$$

- **Known target:** use independent draws from P^q.
- **Observational data:** use a held-out DR IME; its uncertainty remains part of the assessment.

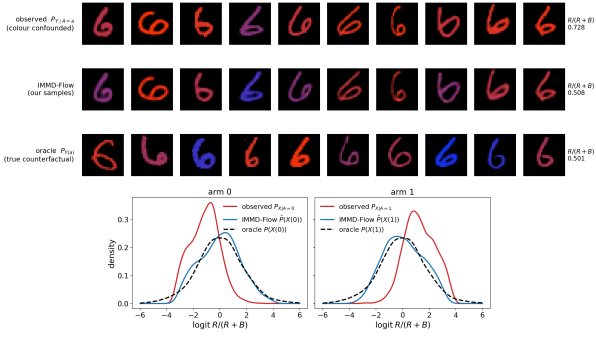
Compare initial, final and baseline particles against the same evaluation target.

The fitted loss $\widehat{F}$ checks optimisation, not target accuracy. Fix the representation as well as the kernel.

Bozkurt et al., submitted to AISTATS 2027.

Transporting image outcomes

Encode observed images, transport the latent particles, then decode. Treatment sets digit identity. Observed colour is confounded.



The transported colour distribution approaches the population target.

IMQ–Euler in fixed encoder coordinates. Decoding and representation accuracy require separate control.

Bozkurt et al., submitted to AISTATS 2027.

Computational cost of particle sampling

Nuisance models estimate the target IME. Particles then optimise directly against that fixed target.

The saving comes from avoiding an additional generator fit.

Fitting + generation (seconds)	DR–IMMD	DecFM plug-in
ColorMNIST	38.4	423.2
CelebA	700.5	8849.5

Total fitting and generation cost is lower in both experiments.

Totals exclude shared preprocessing, decoding and evaluation. DecFM generation alone is faster.

Bozkurt et al., submitted to AISTATS 2027.

IME accuracy matters across applications

The same estimated IMEs support testing and sampling.

- DR estimation: control nuisance bias through a product remainder.
- Global testing: detect distributional changes with calibrated tests.
- Interpretable testing: select informative references and estimate their contrasts efficiently.
- Sampling: match particles to an estimated IME and assess final MMD.

Better IME estimates give more accurate causal contrasts and a better sampling target.

Zenati et al., NeurIPS 2025; Zenati and Gretton, NeurIPS 2026; Bozkurt et al., submitted to AISTATS 2027.

Backup slides

Identification and IME estimation

Confounding · Well-defined score · Orthogonality

Valid probability paths · Asymptotic linearity and efficiency

Distributional testing

MMD degeneracy · Complementary locations

Local power geometry · Regular local alternatives

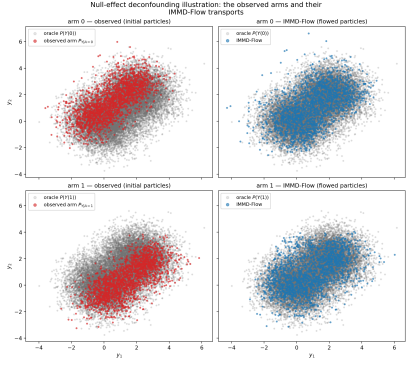
Sampling

Orthogonal optimisation · Kernel regression

DR coefficients · Generated-law guarantee

Causal target: confounding

Observed groups can differ even when the intervention has no effect.



Here $\mathcal{L}_0(Y \mid A = 0) \neq \mathcal{L}_0(Y \mid A = 1)$, but $P^0 = P^1$.

Rows: treatment arms. Left: observed samples (red). Right: generated samples (blue). Grey: the common interventional target.

Bozkurt et al., submitted to AISTATS 2027.

Estimation: well-defined profiles

For $\eta = (m, r)$, the corrected profile is

$$\xi_q(O; \eta) = m_q(X) + r(A, X)\{\phi(Y) - m(A, X)\}.$$

$m_q(x) = \int m(a, x) q(da \mid x)$ averages the candidate CME over the intervention.

Suppose $k(y, y) \le \kappa^2 < \infty$, $r_0, r \in L^2$ and $m \in L^2(\mathcal{H})$. Cauchy–Schwarz then gives

$$\mathbb{E}_0 \|\xi_q(O; \eta)\|_{\mathcal{H}} < \infty.$$

Its population mean exists in $\mathcal{H}$.

This ensures integrability; inference also needs finite variance.

$O = (X, A, Y) \sim P_0$; $\phi(y) = k(y, \cdot)$; e_0: treatment propensity; $r_0 = dq/de_0$ with $q \ll e_0$. $\|r\|_{L^2}^2 = \mathbb{E}_0 |r(A, X)|^2$ and $\|m\|_{L^2(\mathcal{H})}^2 = \mathbb{E}_0 \|m(A, X)\|_{\mathcal{H}}^2$. Standard Borel data spaces, separable $\mathcal{H}$, strongly measurable ϕ.

Zenati, Bozkurt and Gretton, NeurIPS 2025; Bozkurt et al., submitted to AISTATS 2027.

Estimation: Neyman orthogonality

Keep P_0 and q fixed. Perturb the two nuisance functions:

$$\eta_t = (m_0 + t\dot{m},\ r_0 + t\dot{r}), \qquad \eta_0 = (m_0, r_0).$$

Here t is a small real step; $\dot{m}, \dot{r}$ are fixed directions.

The population product-error identity becomes

$$\mathbb{E}_0 \xi_q(O; \eta_t) - \mu_q = -t^2 \mathbb{E}_0[\dot{r}(A, X)\dot{m}(A, X)].$$

$$\frac{d}{dt}\mathbb{E}_0 \xi_q(O; \eta_t)\Big|_{t=0} = 0.$$

The mean corrected profile has no first-order nuisance bias.

ξ_q: corrected profile; $\mathbb{E}_0 \xi_q(O; \eta_0) = \mu_q$. $\dot{m} \in L^2(\mathcal{H})$, $\dot{r} \in L^2$ under the observed (A, X) law. Linear paths are valid in this function space; probability-model paths are covered too.

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

Estimation: valid probability paths

Valid probability paths preserve the same second-order bias.

Treatment assignment: for binary A, perturb its propensity:

$$e_t(1 \mid x) = e_0(1 \mid x) + t\dot{e}(x).$$

$e_0(1 \mid x) \in [\varepsilon, 1-\varepsilon], 0 < \varepsilon < 1/2$; bounded direction $|\dot{e}(x)| \leq 1$; $|t| < \varepsilon/2$. Set $e_t(0 \mid x) = 1 - e_t(1 \mid x)$.

Outcome model: tilt its conditional law by a bounded function s:

$$P_t(dy \mid a, x) = (1 + t\, s(y, a, x))P_0(dy \mid a, x).$$

$\mathbb{E}_0[s(Y, A, X) \mid A, X] = 0$ and $|t\, s(y, a, x)| < 1$ ensure a probability law.

Nuisance changes are $O(t)$; corrected-score bias is $O(t^2)$.

Keep P_0 fixed when evaluating $\mathbb{E}_0\xi_q$; only the nuisance inputs change.

Bozkurt et al., submitted to AISTATS 2027.

Estimation: asymptotic linearity

For independent $O_i \sim P_0$, with cross-fitting and negligible nuisance bias,

$$\sqrt{n}(\widehat{\mu}_q^{\mathrm{DR}} - \mu_q) = \frac{1}{\sqrt{n}} \sum_i \{\xi_q(O_i; \eta_0) - \mu_q\} + o_p(1).$$

Here $\eta_0 = (m_0, r_0)$ contains the true nuisance functions.

Asymptotic linearity: to first order, we average the true corrected profiles.

Their centred version is the canonical gradient: it gives the smallest asymptotic covariance for regular estimation of fixed embedding coordinates.

Evaluating the IME at fixed locations inherits this efficiency.

Foldwise conditions: finite true-score second moment; fitted scores converge in mean square; $\|\widehat{r} - r_0\|_{L^2} \|\widehat{m} - m_0\|_{L^2(\mathcal{H})} = o_p(n^{-1/2})$. The remainder is $o_p(1)$ in $\mathcal{H}$ norm.

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

Testing: degeneracy of squared MMD

With true nuisances, let $d_i = \xi_q(O_i; \eta_0) - \xi_{q'}(O_i; \eta_0)$. Under equal interventional laws, $\mathbb{E}_0 d_i = 0$.

$$U_n = \frac{1}{n(n-1)} \sum_{i \neq j} \langle d_i, d_j \rangle_{\mathcal{H}}.$$

Degenerate: the conditional mean given one observation is zero.

$$\mathbb{E}_0[\langle d_i, d_j \rangle_{\mathcal{H}} \mid d_i] = \langle d_i, \mathbb{E}_0 d_j \rangle_{\mathcal{H}} = 0.$$

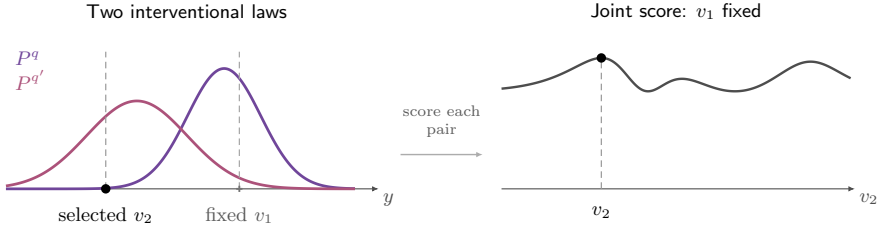
$$nU_n \Rightarrow \sum_{\ell} \lambda_\ell (Z_\ell^2 - 1), \qquad Z_\ell \overset{\text{iid}}{\sim} \mathcal{N}(0,1).$$

λ_ℓ: eigenvalues of the covariance operator of d_i. Independent oracle scores; separable $\mathcal{H}$; finite second moment and nonzero covariance.

Gretton et al., JMLR 2012; Kim and Ramdas, Bernoulli 2024.

Testing: complementary locations

Fix the first reference outcome v_1. Choose v_2 by scoring the pair, accounting for noise *and* correlation.



The second comparison adds information beyond the first.

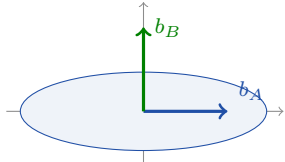
Joint score: $\mathbf{w}(V)^\top (\Sigma(V) + \tau I_2)^{-1} \mathbf{w}(V)$, with $V = (v_1, v_2)$ and $\tau = 10^{-4}$.
$\mathbf{w}(V) = (\mu_q(v_j) - \mu_{q'}(v_j))_{j=1}^2$; $\Sigma(V)$: covariance of the two true corrected similarity differences; I_2: identity. Population illustration.

Zenati and Gretton, NeurIPS 2026.

Backup contents Return to talk

Testing: signal in noise units

Local changes have size $1/\sqrt{n}$, the scale of sampling noise. At fixed test locations V, suppose $\sqrt{n}\,\mathbf{w}_{P_n}(V) \to b \in \mathbb{R}^J$.



An equal-λ ellipse

Two shifts of equal raw size. The vertical direction is less noisy.

Whitening measures the shift relative to its covariance:

$$\lambda = b^\top \Sigma_0^{-1} b.$$

Larger λ gives greater limiting power, at fixed J and level α.

Here b_B is more detectable than b_A.

P_n: local alternative law; $\mathbf{w}_{P_n}(V)$: its true witness vector. At fixed V, write $b = b(V)$ and $\lambda = \lambda(V)$. $\Sigma_0 = \mathrm{Cov}_0[(d(O)(v_j))_{j=1}^J]$ is positive definite, where $d(O) = \xi_q(O;\eta_0) - \xi_{q'}(O;\eta_0)$ uses true null nuisances. Local-limit assumptions.

Zenati and Gretton, NeurIPS 2026.

Testing: the local-power limit

Fix J test locations V. Take n observations from a regular law path $P_n = P_{c/\sqrt{n}}$ starting at a null law P_0; $c \in \mathbb{R}$ is fixed.

The true witness vector satisfies $\sqrt{n}\,\mathbf{w}_{P_n}(V) \to b$. The efficient DR estimate then obeys

$$\sqrt{n}\,\widehat{\mathbf{w}} \Rightarrow \mathcal{N}(b, \Sigma_0).$$

With a consistent covariance estimate and $\gamma_n \to 0$,

$$T_n = n\widehat{\mathbf{w}}^\top(\widehat{\Sigma} + \gamma_n I_J)^{-1}\widehat{\mathbf{w}} \Rightarrow \chi^2_J(\lambda), \qquad \lambda = b^\top \Sigma_0^{-1} b.$$

Power at level α: $\Pr\{\chi^2_J(\lambda) > \chi^2_{J,1-\alpha}\}$ in the limit.

$b = b(V), \lambda = \lambda(V); \Sigma_0 = \Sigma_{P_0}(V) \succ 0$. The population criterion satisfies $nD_{P_n}(V) \to \lambda(V)$; for a fixed nonzero contrast, $nD_P(V)$ instead diverges.

Regular path: quadratic-mean differentiable within the causal model. Assume the efficient expansion at P_0; I_J: identity; $\chi^2_{J,1-\alpha}$: central chi-square quantile.

Zenati and Gretton, NeurIPS 2026.

Sampling: orthogonality of the forces

For fixed Q and nuisance inputs $\eta = (m, r)$, the witness is

$$f_{Q,\eta}(z) = \mu_Q(z) - \mathbb{E}_0[\xi_q(O;\eta)(z)].$$

For real t, $\eta_t = \eta_0 + t(\dot{m}, \dot{r})$ gives

$$f_{Q,\eta_t}(z) - f_{Q,\eta_0}(z) = t^2 \mathbb{E}_0[\dot{r}\,\dot{m}](z).$$

The witness and transport force have zero nuisance derivative.

This holds at every candidate distribution Q.

P_0, q remain fixed; m: CME; r: weight; η_0: truth. Directions $\dot{m} \in L^2(\mathcal{H})$, $\dot{r} \in L^2$ under the observed (A, X) law; bounded feature derivative transfers the result to $-\nabla_z f$.

Bozkurt et al., submitted to AISTATS 2027.

Sampling: kernel regression in closed form

For n observations, set $S_i = (A_i, X_i)$ and $\bar{z}_i = E(Y_i)$. Use an input kernel k_S on treatment–context pairs.

$$K_{ij} = k_S(S_i, S_j), \qquad \mathbf{k}_s = (k_S(S_1, s), \ldots, k_S(S_n, s))^\top.$$

With ridge $\lambda > 0$, set $B_\lambda = (K + n\lambda I_n)^{-1}$. Kernel regression gives

$$\widehat{m}(s) = \sum_{i=1}^{n} [B_\lambda \mathbf{k}_s]_i \, \phi(\bar{z}_i).$$

Predictions combine the observed outcome features.

Next: combine prediction with the weighted residual.

$s = (a, x)$: a treatment–context pair; I_n: identity; $\phi(z) = k(z, \cdot)$; E: fixed encoder, or the identity.

Bozkurt et al., submitted to AISTATS 2027.

Sampling: the DR coefficients

Same-sample full ridge: define the intervention averages

$$[K_q]_{ij} = \int k_S(S_i, (a, X_j)) q(da \mid X_j), \qquad \widehat{\mathbf{r}} = (\widehat{r}(A_i, X_i))_{i=1}^n.$$

The corrected target is $\widehat{\mu}_q^{\mathrm{DR}} = \sum_i c_i \phi(\bar{z}_i)$, where

$$\mathbf{c} = \frac{1}{n}\left[\underbrace{B_\lambda K_q \mathbf{1}_n}_{\text{prediction}} + \underbrace{(I_n - B_\lambda K)\widehat{\mathbf{r}}}_{\text{residual correction}}\right].$$

Compute these coefficients once; freeze them during generation.

Signed coefficients are not resampling probabilities.

K: input Gram matrix; $B_\lambda = (K + n\lambda I_n)^{-1}$; $\mathbf{1}_n$: all-ones vector; $\mathbf{c} = (c_1, \ldots, c_n)^\top$. Regression definitions.

Bozkurt et al., submitted to AISTATS 2027.

Sampling: three sources of error

For a fixed representation, let $e_n = \|\widehat{\mu}_q^{\mathrm{DR}} - \mu_q\|_{\mathcal{H}}$.

After R updates with M particles, the guarantee separates

$$\begin{aligned} \|\mu_{Q_M^{(R)}} - \mu_q\|_{\mathcal{H}} \lesssim\ & e^{-cR^{1-2\alpha}} && \text{optimisation} \\ & + (1 + R^{\alpha})e_n && \text{target estimation} \\ & + R^{\alpha}\sqrt{\log(R/\delta)/M} && \text{finite particles.} \end{aligned}$$

More iterations reduce the first term but can amplify target error.

DR makes the nuisance bias in e_n second order.

$Q_M^{(R)}$: particle law; $c > 0$: constant; $0 \le \alpha < 1/2$: noise-decay exponent; $0 < \delta < 1$: failure probability; $\lesssim$: up to fixed constants.

Conditional probability at least $1 - \delta$ under smooth noisy updates, gradient dominance and a bounded computed target. Decoding needs further conditions.

Bozkurt et al., submitted to AISTATS 2027.